

Load in the data

```
# The here package can just make finding files nice, don't be put off by it.
crop <- read_csv(here::here("data/crop.csv"), col_types = cols())
head(crop) # Check it has read in properly
```

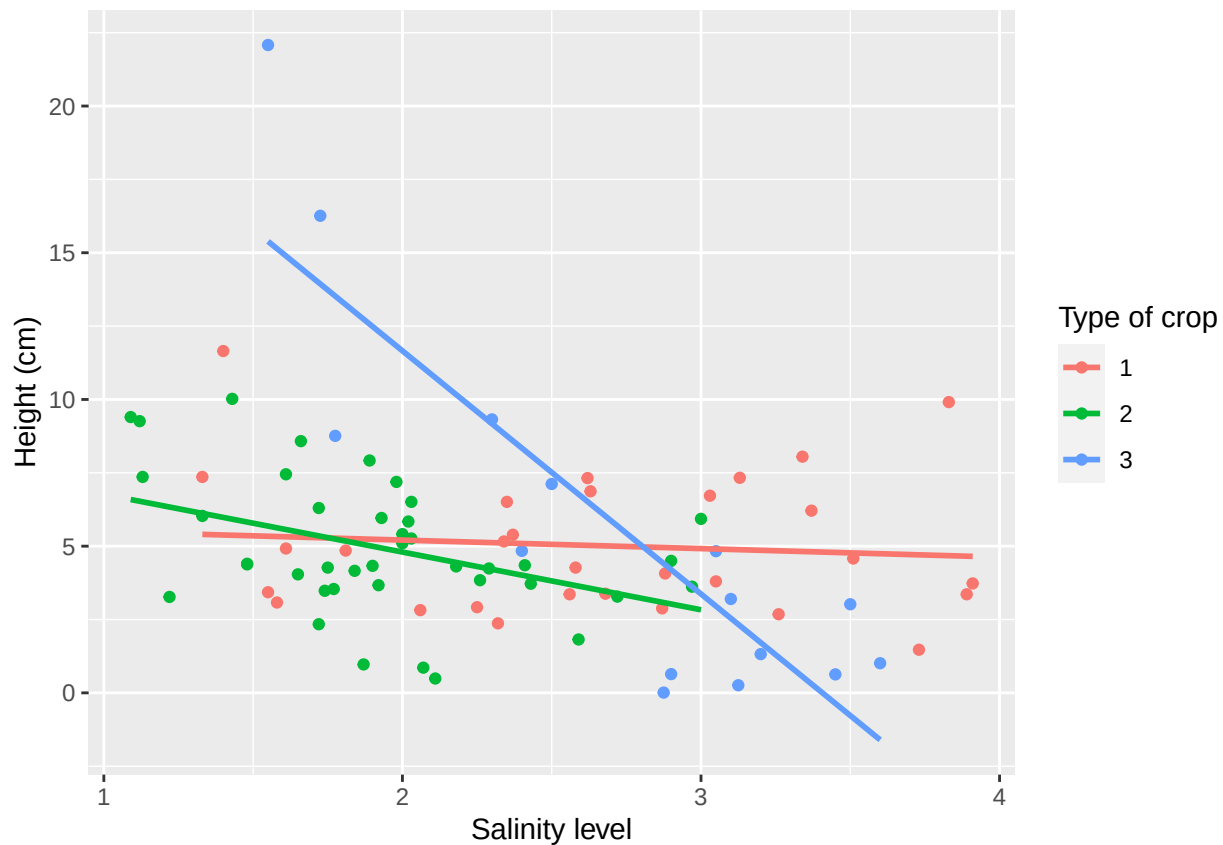
```
## # A tibble: 6 x 3
##   Height Salinity Type
##   <dbl>   <dbl> <dbl>
## 1   6.87     2.63     1
## 2    3.8     3.05     1
## 3    3.43     1.55     1
## 4    3.36     3.89     1
## 5    6.21     3.37     1
## 6    8.05     3.34     1
```

- Type has read in as a double, let's change that to a factor.

```
crop <- crop %>%
  mutate(Type = as.factor(Type))
```

Get a scatter plot

```
crop %>%
  ggplot(aes(x = Salinity, y = Height, colour = Type)) +
  geom_point() +
  geom_smooth(method = lm, formula = y ~ x, se = FALSE) +
  labs(y="Height (cm)", x="Salinity level", col="Type of crop")
```



Fit the 3 different ANCOVA models

```
identical <- lm(Height ~ Salinity, data = crop)
parallel <- lm(Height ~ Salinity + Type, data = crop)
separate <- lm(Height ~ Salinity * Type, data = crop)
```

```
summary(separate)
```

```
##
## Call:
## lm(formula = Height ~ Salinity * Type, data = crop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.7630 -1.6844 -0.1351  1.7698  6.6924
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.7874     1.6536   3.500 0.000768 ***
## Salinity       -0.2902     0.5982  -0.485 0.628884
## Type2          2.9467     2.3131   1.274 0.206433
## Type3         22.4455     3.2230   6.964 8.75e-10 ***
## Salinity:Type2 -1.6772     1.0095  -1.661 0.100612
## Salinity:Type3 -7.9971     1.1517  -6.944 9.56e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 2.449 on 79 degrees of freedom
## Multiple R-squared:  0.4959, Adjusted R-squared:  0.464
## F-statistic: 15.54 on 5 and 79 DF,  p-value: 1.257e-10
```

Test for a significant interaction term

For this, we compare the parallel regressions to the separate regressions using an ANOVA:

```
anova(parallel, separate)
```

```
## Analysis of Variance Table
##
## Model 1: Height ~ Salinity + Type
## Model 2: Height ~ Salinity * Type
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      81 764.76
## 2      79 473.97  2    290.79 24.234 6.206e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

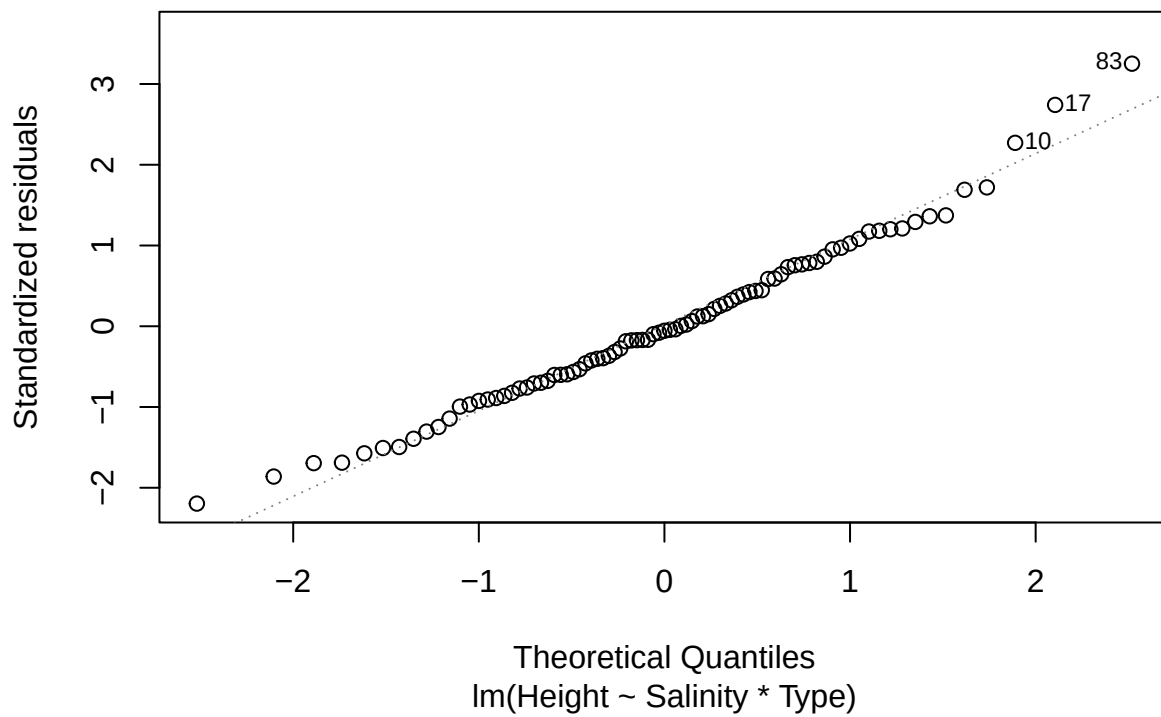
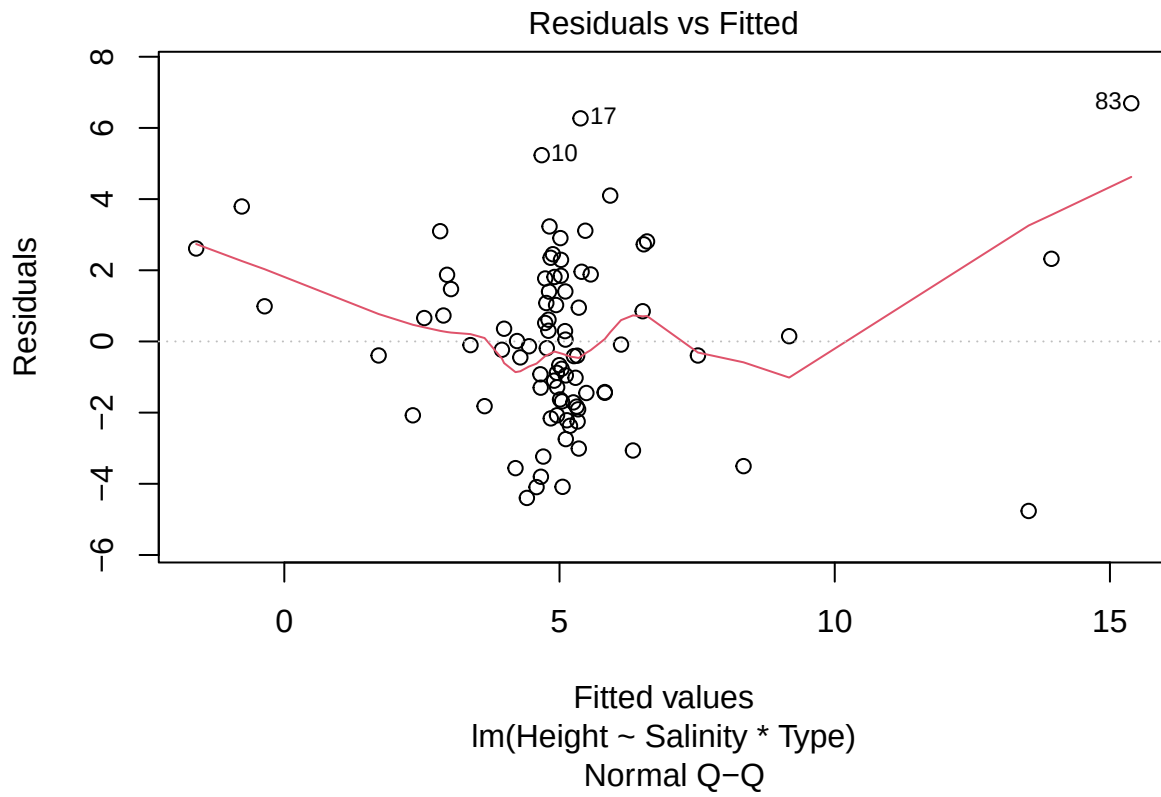
Obtain the BIC

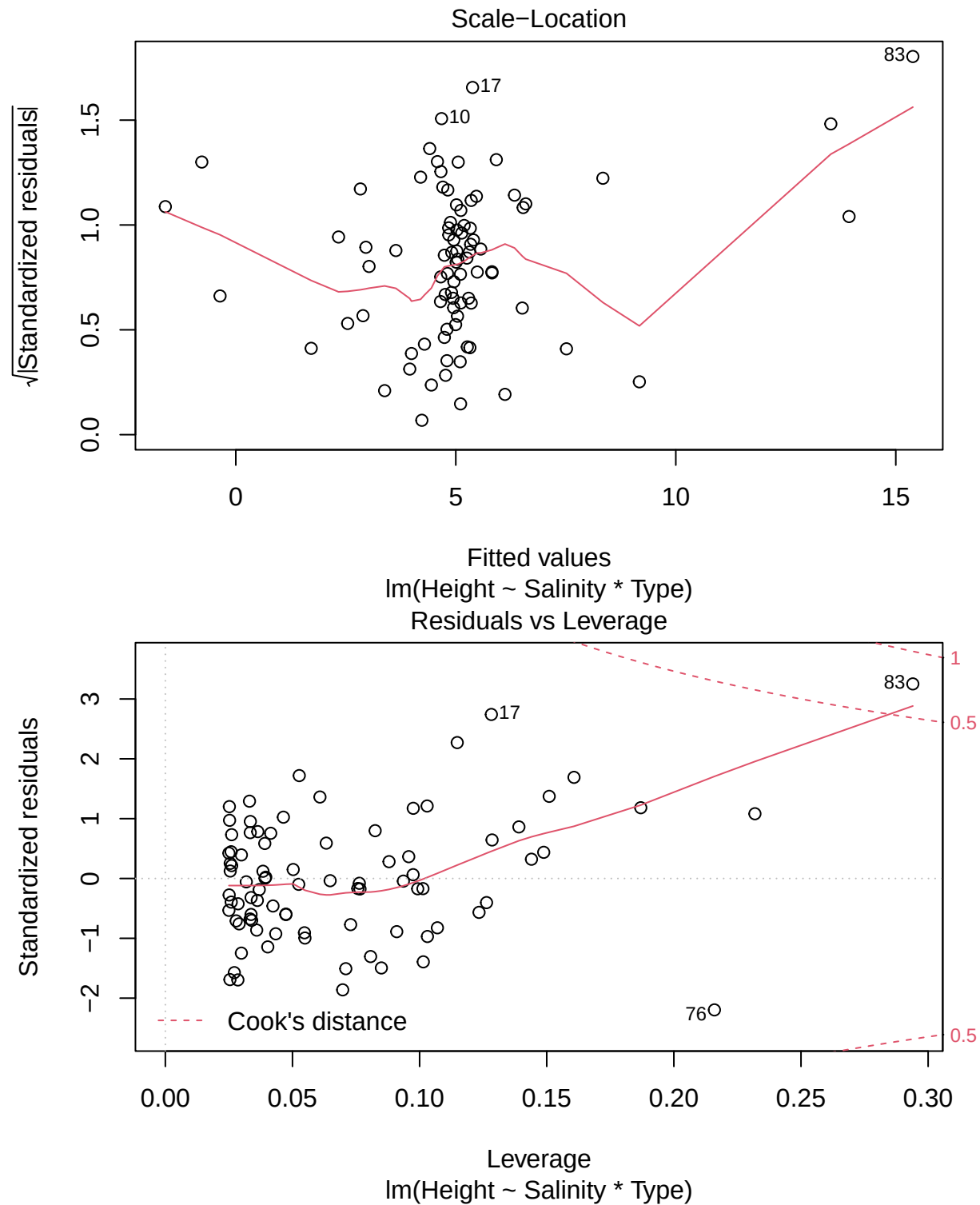
```
BIC(identical, parallel, separate)
```

```
##           df      BIC
## identical  3 448.3398
## parallel   5 450.1701
## separate   7 418.3894
```

Assumption plots:

```
plot(separate)
```





The hard one! contrast time

```
# Load the emmeans library
library(emmeans)
# create emmeans object, we are interested in breaking up the separate model
```

```

# by crop type.
crop_emmeans <- emmeans(separate,~"Type")

## NOTE: Results may be misleading due to involvement in interactions
# Create the contrast object
# This is saying we want to compare type II with type III
crop_contrast <- contrast(crop_emmeans,method=list("Type_II-III"=c(0,1,-1)))
crop_contrast

## contrast      estimate      SE df t.ratio p.value
## Type_II-III    -4.77 0.903 79  -5.280  <.0001

# get the CI
confint(crop_contrast)

## contrast      estimate      SE df lower.CL upper.CL
## Type_II-III    -4.77 0.903 79   -6.56   -2.97
##
## Confidence level used: 0.95

```